Statistical Analysis Report

SSI Improper Payments: Patterns, Causes, and Group Differences (FY2015–2019)

Udacity Data Science Nanodegree — Project 2 | June 2026

1. Overview

This report presents a complete statistical analysis of the **SSI Improper Payments FY2015–2019** dataset, which documents erroneous benefit payments made under the Supplemental Security Income (SSI) program administered by the U.S. Social Security Administration. The dataset is publicly available at https://github.com/qu0est/Data-wrangling-project. The analysis examines the distribution of improper payment amounts, the most frequent error causes, and whether recipient type (disabled adults vs. aged recipients) is associated with differences in payment error magnitude. Understanding the structure of these errors supports efforts to reduce program waste and improve administrative accuracy.

2. Dataset Description

The dataset contains **582 rows and 15 columns**. Each row represents a single SSI improper payment case identified during fiscal years 2015 through 2019. Key variables examined include: **Improper_Payment_Amount_USD** (continuous, range \$38–\$8,476), **Recipient_Type** (categorical: Disabled Adult 18–64, Aged 65+, Disabled Child under 18), **Error_Cause_Category** (categorical, nine categories including Wages/Earned Income and Financial Accounts/Resources), **Duration_of_Error_Months** (continuous, 1–36 months), and **Fiscal_Year** (ordinal, 2015–2019). One column (*Recovery_Action*) contained 32 missing values (5.5%); all other 14 columns were fully complete. Missing recovery action records were retained for all analyses not involving that variable. Overpayments constitute approximately 77% of cases (n=450), with underpayments comprising the remaining 23% (n=132). The mean improper payment amount is **\$3,771.98** (SD = \$2,439.33), and the distribution exhibits mild positive skew (skewness = 0.30).

3. Methods

3.1 Descriptive Statistics

Summary statistics (mean, median, standard deviation, quartiles) were computed for all numeric variables using *pandas* df.describe(). Categorical variable frequencies were tabulated with value_counts(). Reporting both mean and median follows the initial data analysis guidance of Huebner et al. (2024), who recommend characterizing variable distributions before performing any inferential tests to expose skewness, outliers, and potential violations of test assumptions. Skewness was also computed to inform the choice of visualization and test.

3.2 Visualization Choices

Three visual models were created. A **histogram** (Figure 1) was selected to depict the shape and spread of the primary outcome variable (Improper_Payment_Amount_USD), with mean and median reference lines overlaid. A **boxplot** (Figure 2) was used to compare distributions across recipient types because it simultaneously displays median, interquartile range, and outliers for multiple groups (Tukey, 1977). A **grouped bar chart** (Figure 3) was chosen to compare error cause category frequencies across the five fiscal years, as bar charts are the standard method for visualizing count comparisons across categorical groups (Cleveland & McGill, 1984).

3.3 Hypothesis Test

An independent-samples **Welch's t-test** was performed to compare mean improper payment amounts between disabled adults (18–64) and aged recipients (65+). The Welch variant was selected over Student's t-test because it does not assume equal population variances between groups, making it more robust when group sample sizes or variances differ (Zimmermann, 2004). The test was two-tailed at $\alpha = 0.05$. The assumption of approximate normality was assessed by examining skewness; the Central Limit Theorem supports the approximation given group sizes of n■ = 316 and n■ = 183.

4. Results

4.1 Descriptive Statistics

The mean improper payment amount across all cases was **$3,771.98** (Mdn = $3,293.04, SD = $2,439.33). Monthly benefit amounts ranged from $200.63 to $781.81 (M = $500.53). The average duration of error was approximately **17 months** (SD = 10.43). Wages and earned income was the most frequent error cause (n ≈ 105 across all years), followed by financial accounts and resources. Recipient failure to report was the dominant reporting status. Disabled adults (18–64) comprised the largest recipient group (n = 316, 54%).

4.2 Visualizations

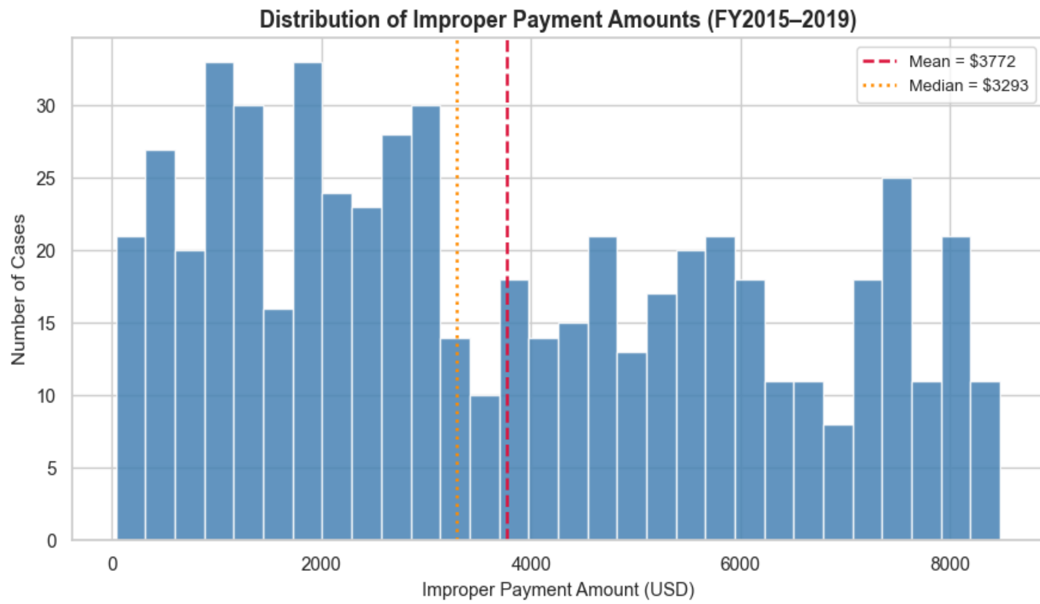

Figure 1. Distribution of improper payment amounts (FY2015–2019). The distribution is right-skewed; the mean ($3,772) exceeds the median ($3,293), indicating a tail of high-value errors.

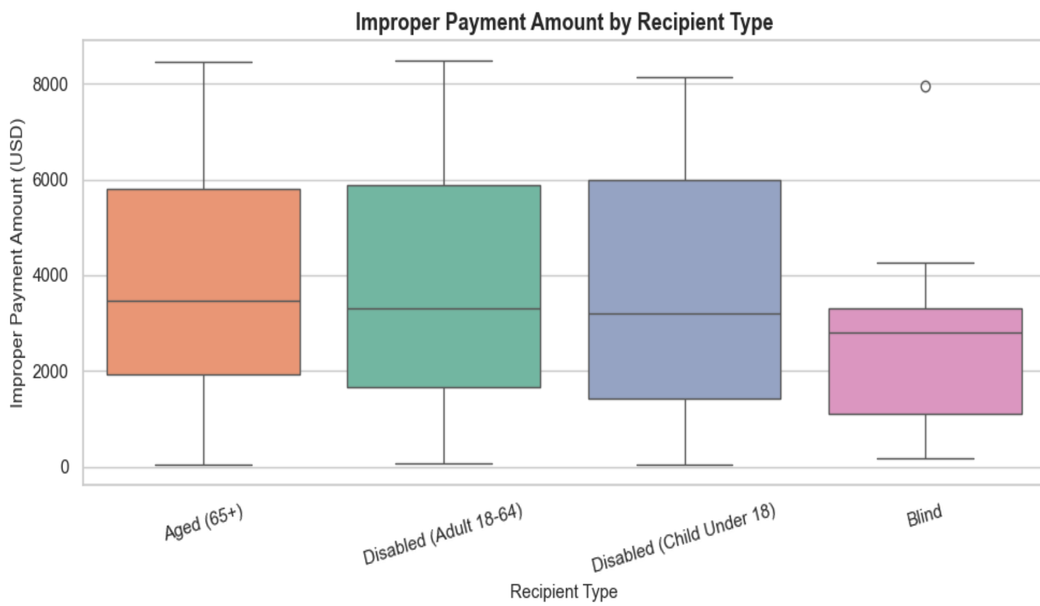

Figure 2. Boxplots of improper payment amounts by recipient type. All three groups show similar medians and overlapping IQRs, with outliers present across all categories.

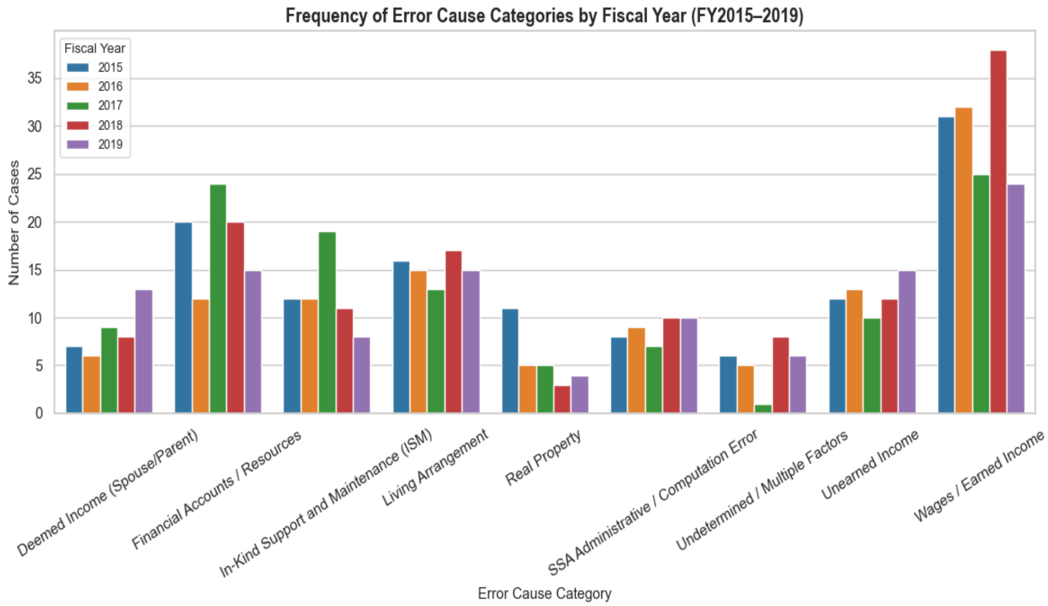

Figure 3. Frequency of error cause categories by fiscal year (FY2015–2019). Wages/Earned Income and Financial Accounts/Resources are the dominant causes consistently across all years.

4.3 Hypothesis Test

Null Hypothesis (H■): The mean improper payment amount for disabled adults (18–64) equals the mean for aged recipients (65+).
Alternative Hypothesis (H■): The mean improper payment amounts differ between the two groups (two-tailed).

Results of the Welch's t-test: **t(490.6) =** –**0.3367, p = 0.7366**. The 95% confidence interval for the mean difference (Disabled – Aged) was [–$514.96, $364.38].

Because p = 0.7366 > α = 0.05, we **fail to reject the null hypothesis**. There is no statistically significant difference in mean improper payment amounts between disabled adults and aged recipients. The confidence interval for the difference spans zero, consistent with this conclusion.

5. Interpretation for a Non-Technical Audience

The Social Security Administration sometimes pays people more or less than they are supposed to receive — these are called improper payments. This analysis looked at 582 such cases recorded between 2015 and 2019. On average, each error involved about $3,772 in incorrect payments, and mistakes typically went undetected for around 17 months. The most common reason errors happen is that recipients do not report changes in their wages or savings in a timely manner, which is a pattern that held steady every year in the dataset. We also asked whether the dollar amount of errors was different depending on whether the recipient was a disabled adult or an elderly person — but the statistical test showed no meaningful difference between those two groups. In plain terms: the size of improper payments is about the same regardless of whether the recipient is younger or older; the real problem appears to be a widespread difficulty in keeping the agency informed of changes in income and assets.

6. Limitations and Potential Bias

6.1 Dataset Limitations

The dataset covers only five fiscal years (2015–2019) and contains 582 cases, which may not be representative of the full population of SSI improper payments. The SSA processes millions of cases annually; the cases in this dataset appear to be a curated or sampled subset, and sampling methodology is not documented. As a result, subgroup analyses (e.g., by state or error cause) have limited statistical power. Additionally, the dataset ends before the COVID-19 pandemic, which substantially disrupted SSA operations, so findings may not generalize to more recent years.

6.2 Potential Bias

A key source of potential bias is **detection bias**: errors are only recorded when they are discovered and documented. Cases involving small amounts or short durations may be systematically under-represented if they fall below audit detection thresholds. This would inflate estimates of mean error amounts and durations. Additionally, the over-representation of "Recipient Failure to Report" as a cause category may reflect an institutional framing bias — cases attributed to SSA administrative errors may be under-coded relative to those attributed to recipients. Analysts should be cautious about using these data to draw causal inferences about recipient behavior (Huebner et al., 2024). Ethical concerns also arise from using this dataset to make policy recommendations about specific demographic groups, as the data do not capture the full social context (language barriers, disability severity, access to banking) that may explain reporting failures.

References

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